M101 · Unit 6

Rates, extrema and the shape of a function

Recall: retrieve readily · Understand: explain conditions · Consult: look up

Use	Topic	Statement and conditions
Recall	**Stationary / critical**	Stationary: $f'(c) = 0$. Interior critical input: $f(c)$ exists and $f'(c) = 0$ or $f'(c)$ does not exist. These are candidates, not classifications.
Understand	**Increasing / decreasing**	On an interval: $f' > 0$ gives strict increase; $f' < 0$ gives strict decrease; $f' \geq 0$ gives nondecrease. Require differentiability inside and continuity at included endpoints. Do not bridge a domain gap.
Recall	**First-derivative test**	At continuous interior candidate c, neighbouring derivative signs $+ \to -$ give a local maximum; $- \to +$ a local minimum; unchanged nonzero signs give neither.
Recall	**Concavity / inflection**	$f'' > 0$: slopes increase, concave up. $f'' < 0$: slopes decrease, concave down. Inflection: change of concavity through a continuous graph point. $f''(c) = 0$ alone is insufficient.
Recall	**Second-derivative test**	$f'(c) = 0$; f differentiable nearby; $f''(c)$ exists. Positive: local minimum; negative: local maximum; zero: inconclusive.
Understand	**Absolute extrema**	Continuous f on closed bounded $[a, b]$ attains maximum and minimum. Compare all interior critical values and both endpoint values. Keep ties. A limiting value need not be attained.
Consult	**Related rates**	Choose a model and signed rates; identify fixed quantities. Differentiate with respect to the common time before inserting instantaneous values. Chain/product rules retain every changing factor. Check units and validity.
Consult	**Rate relations**	$A = ab$: $\dot{A} = b\dot{a} + a\dot{b}$. $d^2 = x^2 + y^2$: $d\dot{d} = x\dot{x} + y\dot{y}$. Fixed-radius sphere: $V = \pi h^2(R - h/3)$, $\dot{V} = \pi h(2R - h)\dot{h}$, $0 < h < 2R$. Constant pV: $V\dot{p} + p\dot{V} = 0$ (absolute pressure).

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Interval reasoning and graph checks

Use	Topic	Statement and conditions
Understand	**Mean value theorem**	Continuous on $[a, b]$, differentiable on (a, b), $a < b$: $$f'(c) = \frac{f(b) - f(a)}{b - a}, \quad a < c < b\,.$$ At least one match; not necessarily unique or at the midpoint.
Recall	**Consequences on an interval**	$f' = 0 \Rightarrow f$ constant; $f' = g' \Rightarrow f - g$ constant. With MVT hypotheses, $m \le f' \le M$ gives $m(b - a) \le f(b) - f(a) \le M(b - a)$.
Understand	**Plot audit**	Function/domain ≠ sample list ≠ displayed line. Record spacing and limits. Separate domain components. Check derivative evidence. Finite sampling cannot generally certify every feature.

Consult · Python quick lookup

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(0, 1, 401)  # endpoints included; 400 gaps
y = x + 0.08 * np.sin(20 * np.pi * x)  # radians
plt.plot(x, y, "o-")       # markers plus connecting segments
plt.xlim(0, 1); plt.ylim(-0.15, 1.15)
plt.xlabel("x (dimensionless)"); plt.ylabel("f(x) (dimensionless)")
plt.show()
# Change n in linspace to change samples; limits change the view.
# For a domain gap, plot the two component arrays separately.
```

Notation: primes refer to the displayed input; dots to time. a, b denote endpoints or panel sides as labelled; c an interior input; R a fixed radius; h depth; p absolute pressure; V volume. A rate has output units per input unit.

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